

Team name:		Kavosh, Simul				Name of the Local Currency		Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
						Brazilian Real		R\$0.00	\$0.00
#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local)	Cost (USA)		
#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local)	Cost (USA)		
1	Python	Programming language	Main programming language used for the robot controller, navigation, mapping support, vision integration, verification, and reporting.	https://www.python.org/	Python Software Foundation	FREE	FREE		
2	Webots / Erebus Simulation	Simulation software / robot API	Simulation environment and robot/sensor interface used for RoboCupJunior Rescue Simulation controller development and testing.	https://cyberbotics.com/	Cyberbotics / RoboCupJunior Erebus	FREE	FREE		
3	OpenCV	Computer vision library	Used for ROI processing, HSV colour segmentation, contour filtering, circularity checks, ring sampling, annotation overlay review, and cognitive target detection.	https://opencv.org/	OpenCV.org	FREE	FREE		
4	Ultralytics YOLO11s	AI model architecture	Object-detection architecture used as the base for the team-trained Greek victim-symbol detector.	https://docs.ultralytics.com/models/yolo11/	Ultralytics	FREE	FREE		
5	Team-Trained YOLO11s Victim Model	Custom AI model	Team-trained model used to detect Greek victim symbols Φ, Ψ, and Ω before converting them to H, S, and U report codes.	Team-developed	Kavosh, Simul / Ernia Karimi	FREE	FREE		
6	ROI-Based Wall Candidate Pipeline	Custom vision algorithm	Filters side-camera frames for meaningful wall candidates, crops the region of interest, and sends only the ROI to YOLO11s or OpenCV.	Team-developed	Kavosh, Simul / Ernia Karimi	FREE	FREE		
7	OpenCV Cognitive Target Detector	Custom computer-vision algorithm	Rule-based detector for cognitive targets using HSV segmentation, circular contour filtering, five-ring sampling, colour classification, and numerical target evaluation.	Team-developed using OpenCV	Kavosh, Simul / Ernia Karimi	FREE	FREE		
8	YOLO Annotation QA Overlay Tool	Custom dataset QA tool	Uses OpenCV to render YOLO-format labels and bounding boxes onto images for annotation review and correction.	Team-developed using OpenCV	Kavosh, Simul / Ernia Karimi	FREE	FREE		
9	Hard-Negative Dataset Mining	Custom dataset-engineering process	Process used to collect empty walls, corners, shadows, low-contrast surfaces, maze artifacts, and confusing patterns to reduce false positives.	Team-developed	Kavosh, Simul / Ernia Karimi	FREE	FREE		
10	Temporal Confirmation Logic	Custom verification algorithm	Requires agreement across repeated observations before accepting a wall-token detection, reducing unstable single-frame reports.	Team-developed	Kavosh, Simul / Ernia Karimi	FREE	FREE		
11	LiDAR-Based Fake-Victim Rejection	Custom verification algorithm	Uses LiDAR depth consistency against expected wall geometry to reject protruding 3D fake victims before reporting.	Team-developed	Kavosh, Simul / Ernia Karimi	FREE	FREE		
12	Location-Based Duplicate Suppression	Custom reporting algorithm	Estimates physical wall-token position using GPS, yaw, camera side, and LiDAR distance to prevent repeated reports of the same token.	Team-developed	Kavosh, Simul / Sam Liu and Ernia Karimi	FREE	FREE		
13	A*-Style Grid Path Planning	Custom navigation algorithm	Grid-based path planner using 8-direction neighbour search, movement cost, turn cost, unsafe-cell avoidance, and reachable-target selection.	Team-developed	Kavosh, Simul / Hanwen Yang and Zhengyu Wu	FREE	FREE		
14	PID Movement Correction	Custom control algorithm	Used for waypoint following, yaw correction, wheel-speed correction, and stable movement during navigation.	Team-developed	Kavosh, Simul / Hanwen Yang and Zhengyu Wu	FREE	FREE		
15	LiDAR Wall Alignment and Obstacle Logic	Custom navigation algorithm	Uses LiDAR scan data for obstacle detection, wall alignment, blocked-direction detection, and movement recovery.	Team-developed	Kavosh, Simul / Hanwen Yang and Zhengyu Wu	FREE	FREE		
16	Foresight Hole Detection Logic	Custom safety algorithm	Uses the downward-facing distance sensor only for early hole detection, unsafe-cell marking, backing away, and re-planning.	Team-developed	Kavosh, Simul / Hanwen Yang and Zhengyu Wu	FREE	FREE		
17	Tile-Based Grid Mapping	Custom mapping algorithm	Stores tile status, tile type, visit count, wall data, unsafe holes, reported targets, inferred nearby features, and wall-side inspection state.	Team-developed	Kavosh, Simul / Sam Liu	FREE	FREE		
18	LiDAR Pattern Matching for Wall Classification	Custom mapping algorithm	Compares LiDAR scan patterns to classify straight walls, corners, openings, blocked directions, and wall-adjacent structures for mapping accuracy.	Team-developed	Kavosh, Simul / Sam Liu	FREE	FREE		
19	Lightweight Python SLAM / Map Support	Custom mapping system	Python-based SLAM/map-support direction without ROS, using GPS, yaw, LiDAR, movement data, and side-camera wall observations.	Team-developed	Kavosh, Simul / Sam Liu	FREE	FREE		
20	Wall-Focused Exploration Strategy	Custom planning algorithm	Uses map/SLAM wall data to prioritize uninspected wall sections and prefer wall-adjacent routes for side-camera token detection.	Team-developed	Kavosh, Simul / Sam Liu, Hanwen Yang, and Zhengyu Wu	FREE	FREE		
21	Erebus Reporting Interface Logic	Custom reporting logic	Prepares and sends confirmed victim/report data after verification, duplicate suppression, and estimated wall-token position calculation.	Team-developed for Webots/Erebus controller	Kavosh, Simul	FREE	FREE		
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